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CME 313
MW# 9
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Repeat using 3500 mg BaCl₂, same A_w, but A_s = 1.00 × 10⁻³ m/s

$$3500 \text{ mg BaCl}_2 = 3.500 \text{ Kg BaCl}_2 \cdot \frac{\text{kmol}}{208.23 \text{ Kg}} \cdot \frac{1000 \text{ mol}}{\text{kmol}} = 16.808 \text{ mol BaCl}_2$$

for water: $\frac{996 \text{ Kg H}_2\text{O}}{\text{m}^3}$

$$\frac{16.808 \text{ mol BaCl}_2/\text{m}^3}{996 \text{ Kg H}_2\text{O}/\text{m}^3} = 0.01688 \text{ mol BaCl}_2/\text{Kg H}_2\text{O}$$

$\pi_1 = 0.7825 \text{ atm}$ from table

for the other stream assume $c_2 = 0.1 \text{ Kg BaCl}_2/\text{m}^3$

$$\frac{0.1 \text{ Kg BaCl}_2/\text{m}^3}{996.9 \text{ Kg H}_2\text{O}/\text{m}^3} \cdot \frac{\text{kmol}}{208.23 \text{ Kg}} \cdot \frac{1000 \text{ mol}}{\text{kg}} = 4.822 \times 10^{-4} \text{ mol BaCl}_2/\text{Kg H}_2\text{O}$$

From the table: $\pi_2 = 0.02266 \text{ atm}$

$$\Delta\pi = 0.7825 \text{ atm} - 0.02266 \text{ atm} = 0.782 \text{ atm}$$

$$N_w = 3.50 \times 10^{-9} \frac{\text{Kg}}{\text{cm}^2 \cdot \text{min}} (35.50 \text{ atm} - 0.782 \text{ atm}) = \boxed{0.01215 \frac{\text{Kg}}{\text{s} \cdot \text{m}^2}}$$

$$B = A_w = \frac{3.50 \times 10^{-9} \text{ Kg}/\text{s} \cdot \text{m}^2 \cdot \text{atm}}{A_w c_w (1.00 \times 10^{-3} \text{ m/s})(997 \text{ Kg}/\text{m}^3)} = 3.51 \text{ atm}^{-1}$$

$$R = \frac{B(\Delta P - \Delta\pi)}{1 + B(\Delta P - \Delta\pi)} = \frac{3.51 \text{ atm}^{-1} (35.5 \text{ atm} - 0.782 \text{ atm})}{1 + 3.51 \text{ atm}^{-1} (35.5 \text{ atm} - 0.782 \text{ atm})} = \boxed{0.992}$$

$$C_2 = C_1 - R_{C1}$$

$$C_2 = \left(3.50 \frac{\text{kg}}{\text{m}^3} \right) - \left(3.50 \frac{\text{kg}}{\text{m}^3} \right) (0.992) = \boxed{0.0285 \frac{\text{kg}}{\text{m}^3}}$$

$$N_s = A_s (C_1 - C_2) = (1 \times 10^{-7} \text{ m}^2) (3.50 \frac{\text{kg}}{\text{m}^3} - 0.0285 \frac{\text{kg}}{\text{m}^3}) = \boxed{3.47 \times 10^{-7} \frac{\text{kg}}{\text{m}^2 \cdot \text{s}}}$$

29.7-1) Diffusion Through Liquids and a Membrane

recovered by dialysis

$$C_1 = 2.0 \times 10^{-2} \text{ kmol A/m}^3$$

$$C_2 = 0.8 \times 10^{-2} \text{ kmol A/m}^3$$

$$L = 1.59 \times 10^{-5} \text{ m}$$

$$K' = 0.75$$

$$D_{AB} = 3.5 \times 10^{-11} \text{ m}^2/\text{s} \text{ in the membrane}$$

$$K_{C1} = 3.5 \times 10^{-5} \text{ m/s}$$

$$K_{C2} = 2.1 \times 10^{-5} \text{ m/s}$$

Calculate the individual resistances, total resistance, and total percent resistance of the two films.

Call the resistances R_1 , R_{12} , R_2

$$R_1 = 1/K_{C1} = 1/3.5 \times 10^{-5} \text{ m/s} = \boxed{2.857 \times 10^4 \text{ s/m}}$$

$$R_{12} = 1/k_m = 1/(D_{AB} K'/L) = L/D_{AB} K' = 1.59 \times 10^{-5} \text{ m} / (3.5 \times 10^{-11} \text{ m}^2/\text{s} \cdot 0.75)$$

$$R_{12} = \boxed{6.057 \times 10^5 \text{ s/m}}$$

$$R_2 = 1/K_{C2} = 1/2.1 \times 10^{-5} \text{ m/s} = \boxed{4.762 \times 10^4 \text{ s/m}}$$

$$R_{\text{tot}} = R_1 + R_{12} + R_2 = (2.857 \times 10^4 + 6.057 \times 10^5 + 4.762 \times 10^4) \text{ s/m}$$

$$R_{\text{tot}} = \boxed{6.82 \times 10^5 \text{ s/m}}$$

For the product, assume $c_2 = 0.1 \text{ kg NaCl/m}^3$. Since the solution is dilute, assume $\rho = 997 \text{ kg/m}^3$

So the concentration of water is $(997 - 0.1) \text{ kg/m}^3$

$$\text{So: } \frac{0.1 \text{ kg NaCl/m}^3}{996.9 \text{ kg H}_2\text{O/m}^3} \times \frac{\text{Kmol}}{58.94 \text{ kg}} \times \frac{1000 \text{ mol}}{\text{Kmol}} = 0.001716 \frac{\text{mol NaCl}}{\text{kg H}_2\text{O}}$$

By interpolation

$$\pi_2 = 0.0807 \text{ atm}$$

So

$$N_w = 3.50 \times 10^{-9} \frac{\text{kg}}{\text{s} \cdot \text{m}^2 \cdot \text{atm}} (35.50 \text{ atm} - (2.748 - 0.0807 \text{ atm}))$$

$$N_w = 0.01149 \frac{\text{kg}}{\text{s} \cdot \text{m}^2}$$

part b ?

$$R = \frac{B(\Delta P - \Delta \pi)}{1 + B(\Delta P - \Delta \pi)}$$

$$\text{for } B \text{ use } B = \frac{A_w}{A_{sw}} = \frac{3.50 \times 10^{-9} \text{ kg/s} \cdot \text{m}^2 \cdot \text{atm}}{(2.50 \times 10^{-7} \text{ m}^2/\text{s})(997 \text{ kg/m}^3)} = 1.409 \text{ atm}^{-1}$$

So:

$$R = \frac{1.409 \text{ atm}^{-1} (35.50 \text{ atm} - 2.668 \text{ atm})}{1 + 1.409 \text{ atm}^{-1} (35.50 \text{ atm} - 2.668 \text{ atm})} = 0.9788$$

$$R = \frac{C_1 - C_2}{C_1} \therefore C_2 = C_1 - RC_1$$

$$C_2 = 3.50 \text{ kg/m}^3 - (3.50 \text{ kg/m}^3)(0.9788) = 0.0743 \text{ kg/m}^3 = 74.3 \text{ mg/L}$$

$$N_s = A_s(C_1 - C_2) = (2.50 \times 10^{-7} \text{ m}^2/\text{s}) \left(3.5 \frac{\text{kg}}{\text{m}^3} - 0.0743 \frac{\text{kg}}{\text{m}^3} \right) = 8.56 \times 10^{-7} \frac{\text{kg}}{\text{s}}$$